😀 [Latest] Wiring Tutorial for Xiaozhi AI Breadboard with Camera Interactive Function

First, I would like to thank Boss Zhou Ke for his technical support and for providing the solution.

1. Hardware Required for DIY

- Development Board: ESP32-S3 N16R8 CAM Development Board with OV3660 Camera
- Digital Microphone: INMP441 or ICS43434 Module
- Power Amplifier: MAX98357A
- Cavity speaker: 8Ω 2~3W or 4Ω 2~3W
- Wires: One box of jumpers, several DuPont wires
- 2 pieces of 400-hole breadboards
- SPI-LCD Color Screen (This article takes the 1.54-inch screen as an example; for other screens, please select and compile the corresponding firmware yourself)

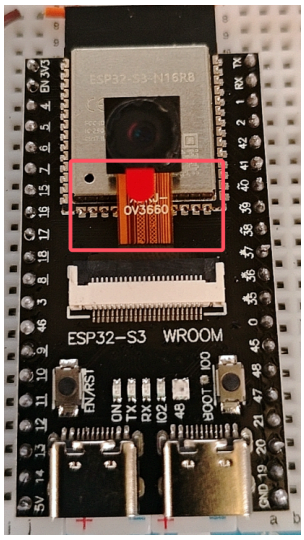
In addition, you may also need a soldering iron, a Type-C data cable, and a computer for firmware flashing.

Recommended purchase link for material package(Pre-soldered pin headers, easy to use without pitfalls, please support)

【Taobao】7-Day No-Reason Return https://e.tb.cn/h.7EOIBStikhYGfQf?tk=1opoUOMtDwm HU006 「Officially Recommended」Xiaozhi AI with Camera for Item Recognition, Lingo, Song Listening, Material Package, Finished AI Dialogue」
Click the link to open directly or search on Taobao to open directly

1.1 ESP32-S3 N16R8 CAM Development Board with OV3660 Camera

Those with poorly soldered pin headers need to solder them properly on their own, connect the camera, with the ov3660 logo facing forward, open the buckle, insert, and press firmly.
(Note: Do not connect while powered on, as the camera is prone to burning out)



1.2 SPI-LCD Color Screen (with Driver Board, Soldered 8-Pin Header)

This tutorial takes a 1.54-inch LCD color screen as an example

2. Wiring Scheme

2.1 Wiring Comparison Table for Development Board and Component Modules

Note: Please wire according to the pins in the table. The wiring photos corresponding to the development board are for reference only.

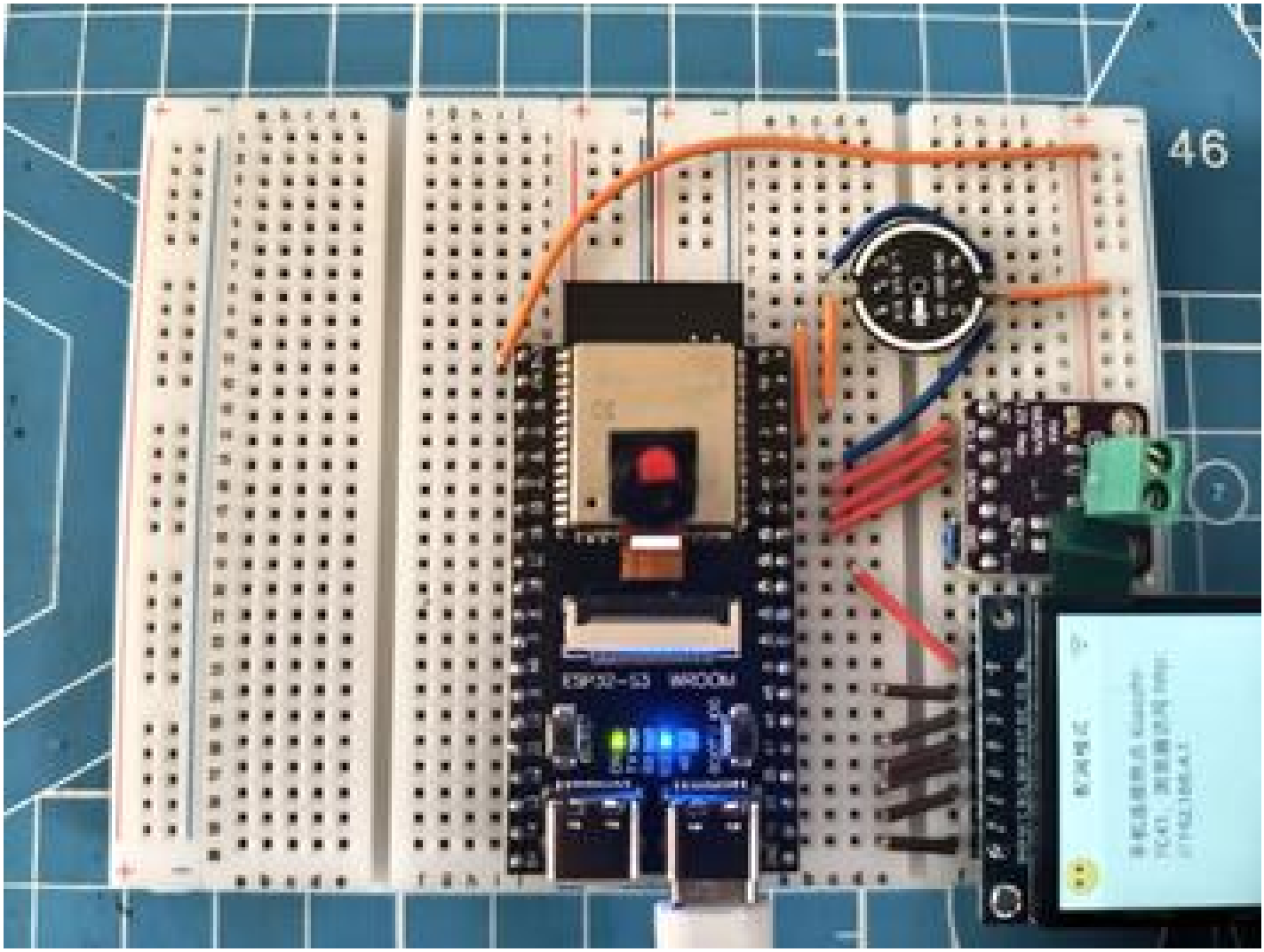
- Connect the ESP32S3 development board 's **bolded numbers/letters** **pins** , and microphone, digital power amplifier, LCD display etc. **bolded English letters** **pins** , **with wires, short-circuit means also connecting wires together** .

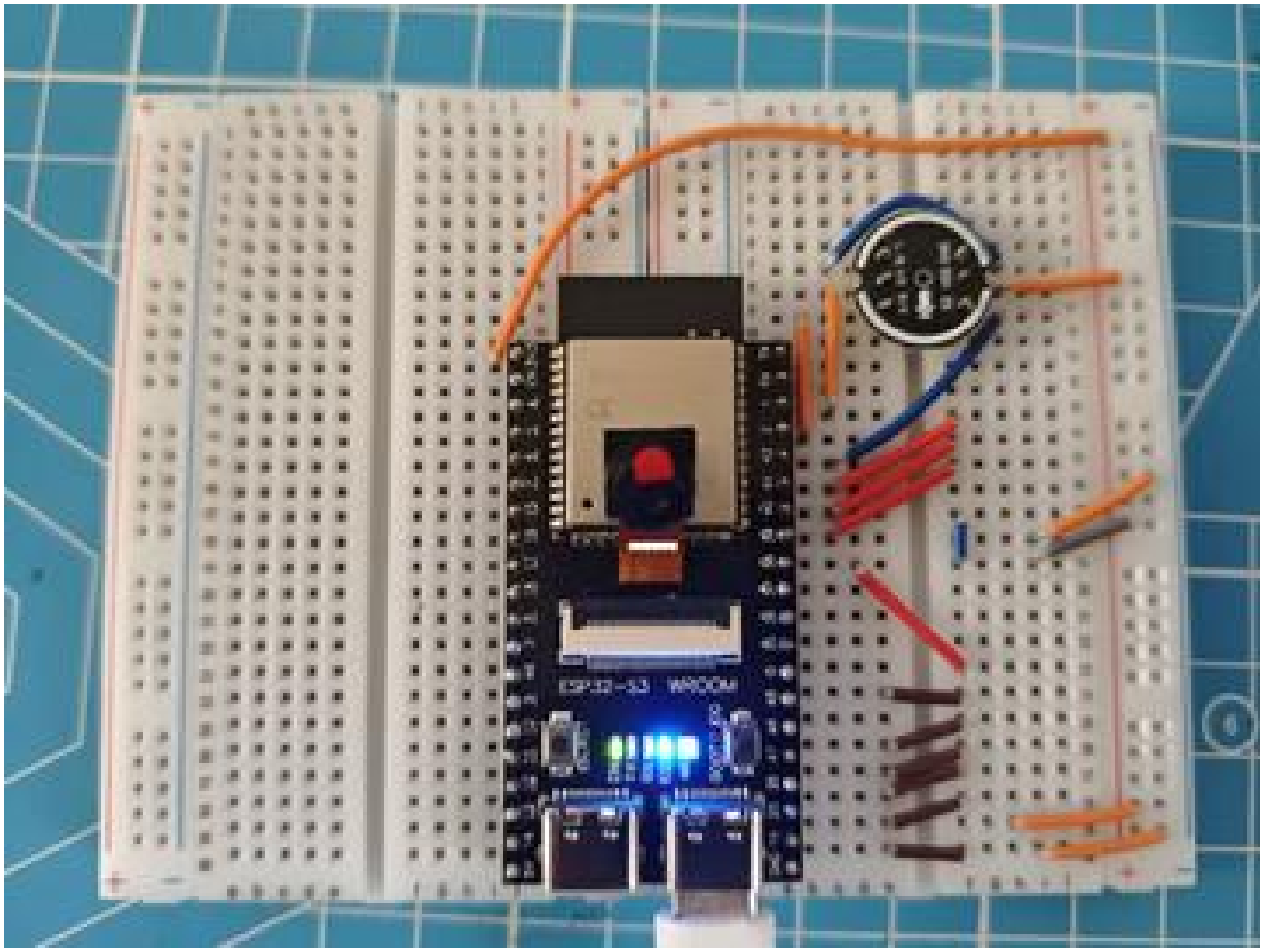
If the wiring is difficult to connect, you can use DuPont wire (wire with a needle) to first pass through the hole, and then connect the conductor

ESP32S3 Development Board	**Microphone INMP441 (I2S Interface)**
GPIO**1**	**WS** Data Selection
GPIO**2**	**SCK** Data Clock
GPIO**42**	**SD** Data Output
3V3	**VDD** Positive power supply 3.3V
GND	**GND** Ground **Short Circuit** **L/R** Left/Right Channel
	Digital Power Amplifier MAX98357A
GPIO**39**	**DIN** Digital Signal
GPIO**40**	**BCLK** Bit Clock
GPIO**41**	**LRC** Left/Right Clock
3V3 /3.3V	**Vin (or VCC) Power Input Short Circuit SD** Shutdown Channel
GND	**GND** Ground
	Audio + Connect **to the positive terminal of the speaker** (**generally, positive and negative terminals are not distinguished**)
	Audio - Connect **to Speaker Negative**
	Display (SPI-LCD 8Pin Interface)
GND	**GND** Ground
3V3	**VCC** Positive Power Supply
GPIO**19**	**SCL** Clock Line
GPIO**20**	**SDA** Data Signal
GPIO**21**	**RST** Reset
GPIO**47**	**DC** Data Selection
GPIO**45**	**CS**Chip Select

GPIO**38**	**BLK** Backlight

After wiring is completed, it is as shown in the figure





Comes with a wiring video tutorial

Note: The main board must be fully pressed down to fit tightly against the breadboard. Often, people leave half of the main board pins exposed and then ask why it doesn't work. If it can't be pressed down, you can first use a DuPont wire to clear the holes.

3. Firmware Download

v2.0.3-bread-compact-wifi-s3cam-3660.bin
9.16MB

Flashing Tutorial:

Flash Tool/Web-based Firmware Flashing (without IDF Development Environment)
One-click Online Flashing: https://sl.ltil.xyz/esp32s3cam/

For programming, insert into the development board**the hole on the right**

Systems below Win11 may not have drivers installed; please install the drivers first

CH340 Series:

https://www.wch.cn/products/CH340.html

CH343 Series:

https://www.wch.cn/products/CH343.html

Operation Tutorial:

Xiaozhi Operation Instructions